

Original Article

Energy-Aware Cluster-Based Routing Protocol for Heterogeneous Wireless Sensor Networks

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Abstract – Wireless Sensor Networks (WSNs) deployed in precision agriculture, environmental monitoring, industrial surveillance, and smart infrastructure increasingly comprise nodes with heterogeneous initial energy capacities. Classical cluster-based protocols such as LEACH and SEP assume homogeneous node populations and consequently fail to exploit the energy surplus of higher-capacity advanced and super nodes, leading to premature hotspot depletion and shortened network lifetime. This paper proposes the Energy-Aware Cluster-Based Routing Protocol for Heterogeneous WSNs (EA-CBRHP), a novel multi-level clustering and multi-hop inter-cluster routing scheme that explicitly accounts for node heterogeneity in both cluster head election and intra-cluster communication. The EA-CBRHP threshold function weights residual energy, initial energy ratio, and node type to elect cluster heads preferentially from energy-rich advanced and super nodes, while a multi-hop inter-cluster routing layer reduces long-distance transmission costs by relaying data through intermediary cluster heads.

Keywords – *Wireless Sensor Networks, Cluster-Based Routing, Heterogeneous WSN, Energy-Aware Protocol, LEACH, Cluster Head Election.*

1. INTRODUCTION:

Wireless Sensor Networks consist of spatially distributed autonomous devices that sense physical or environmental conditions and cooperatively transmit the collected data through the network to a base station or sink. Since their conceptualization in the late 1990s, WSNs have found deployment across a remarkable breadth of application domains: monitoring soil temperature and moisture in precision agriculture, detecting structural fatigue in bridges and dams, tracking animal movements in ecological reserves, measuring air quality and particulate concentrations in urban environments, and providing early warning of industrial equipment failure through vibration and acoustic emission sensing. The unifying constraint across all these applications is energy: sensor nodes are typically battery-powered, physically inaccessible for maintenance, and expected to operate continuously for months or years without human intervention.

The energy challenge in WSNs is primarily a routing challenge. Each packet generated by a sensor node

must traverse a series of wireless links to reach the base station; the energy consumed per bit-metre of transmission scales with distance raised to a power of 2 (free-space model) or 4 (multipath fading model), making long-distance direct transmission disproportionately expensive. Cluster-based routing in which nodes self-organize into groups, elect a cluster head (CH) to aggregate and forward data, and the CH communicates with the base station is the dominant architectural paradigm for energy-efficient WSN routing. By confining intra-cluster communication to short distances and concentrating long-distance transmission in a small number of cluster heads, clustering dramatically reduces network-wide energy expenditure compared to direct transmission or flat routing.

The seminal Low-Energy Adaptive Clustering Hierarchy (LEACH) protocol established the cluster-based routing paradigm in 2000. LEACH's randomized CH rotation and TDMA scheduling within clusters delivered substantial energy savings over naïve direct transmission. However, LEACH and its successors make a critical simplifying assumption: all sensor nodes begin with identical initial energy. In real WSN deployments, this homogeneity assumption is rarely valid. Deploying nodes with different battery capacities (e.g., standard AA versus lithium primary cells), supplementing some nodes with solar energy harvesting, or replacing depleted nodes with fresh units all produce a heterogeneous energy landscape. When LEACH elects cluster heads by random rotation without energy weighting, it imposes equal CH duty on both full-capacity advanced nodes and near-depleted normal nodes, consistently over-stressing the energy-poor nodes and causing premature, spatially correlated node death the routing hotspot problem.

The Stable Election Protocol (SEP) and its descendants address this by defining two node types (normal and advanced) and assigning higher CH election probability to advanced nodes proportional to their extra energy. While a significant improvement, SEP's binary taxonomy cannot adequately represent real-world multi-level heterogeneity (e.g., nodes with solar panels, nodes with larger battery packs, and freshly replaced nodes coexisting in the same field), nor does it account for residual energy at run time an advanced node that has served many CH rounds may have less residual energy than a normal node that has never been elected.

This paper addresses these limitations through the EA-CBRHP protocol. The key innovations are: (i) a three-level node taxonomy (normal, advanced, super) with energy ratios α and β respectively; (ii) a heterogeneity-aware threshold function that weights both initial energy ratio and current residual energy in CH probability, ensuring that CH duty tracks actual energy availability rather than nominal node class; (iii) a multi-hop inter-cluster backbone layer in which distant cluster heads relay data through

geographically intermediate cluster heads rather than transmitting directly to the base station, eliminating the long-distance single-hop CH-to-BS transmission that dominates energy consumption in large fields; and (iv) a TDMA-based intra-cluster scheduling with adaptive slot assignment weighted by node data generation rate, eliminating idle listening overhead within clusters.

2. RELATED WORK:

2.1 Classical Cluster-Based Protocols

Heinzelman et al. (2000) introduced LEACH, the first self-organising cluster protocol for WSNs, demonstrating 8× energy improvement over direct transmission on a 100-node homogeneous network. LEACH's random CH rotation, TDMA intra-cluster scheduling, and data aggregation at the CH set the template for a large subsequent literature. LEACH-C (Heinzelman et al., 2002) moved CH selection to a centralised algorithm at the base station that solves an approximately optimal CH placement problem, reducing unnecessary CH rotations but requiring global network state at the BS and incurring communication overhead proportional to N each round. Younis and Fahmy (2004) proposed HEED, in which CH candidates iteratively announce their candidacy with costs proportional to their residual energy and intra-cluster communication cost, producing a distributed CH election that converges in $O(1)$ iterations without requiring time synchronisation. HEED represents the most competitive prior baseline for EA-CBRHP, and the performance improvement over HEED is a primary evaluation focus of this work.

2.2 Heterogeneity-Aware Protocols

Smaragdakis et al. (2004) proposed SEP (Stable Election Protocol), extending LEACH to a two-level heterogeneous network by assigning advanced nodes an election probability $(1+\alpha)$ times higher than normal nodes, where α is the extra energy fraction of advanced nodes. SEP demonstrated that network stability (FND round) improves significantly when advanced nodes take a disproportionate share of CH duty. Qing et al. (2006) generalised SEP to multi-level heterogeneity in DEEC (Distributed Energy-Efficient Clustering), assigning CH probability proportional to each node's current-to-average energy ratio. DEEC outperforms SEP by tracking residual energy dynamics but its routing layer remains single-hop CH-to-BS. Elbhiri et al. (2010) proposed DDEEC, which modifies DEEC's threshold function to assign reduced probability to advanced nodes after they reach the energy level of normal nodes, preventing over-election of advanced nodes that have been depleted through past CH service. These protocols all share a common limitation: their CH-to-BS routing is single-hop, making them energy-inefficient when the BS is far from the sensing field or the network radius is large.

2.3 Multi-Hop and Hierarchical Routing

Lindsey and Raghavendra (2002) proposed PEGASIS (Power-Efficient GATHERing in Sensor Information Systems), which chains all nodes into a linear sequence and rotates the transmission duty along the chain, achieving better load balance than LEACH but incurring high chain-construction overhead and poor fault tolerance. Manjeshwar and Agrawal (2001) introduced TEEN (Threshold-sensitive Energy Efficient sensor Network protocol), in which nodes transmit only when sensed values cross a user-defined hard threshold, dramatically reducing transmission frequency for slowly changing environments. TEEN's energy savings are highly application-dependent and its latency is unpredictable. Muruganathan et al. (2005) proposed a centralized LEACH variant with multi-hop inter-cluster routing that routes CH data through intermediate CHs rather than direct-to-BS, demonstrating 30–40% lifetime improvement for network radii above 100 m — a result that motivates the multi-hop inter-cluster backbone in EA-CBRHP. Recent work by Logambigai and Kannan (2016) proposed FLECH, using fuzzy logic for CH selection in heterogeneous networks, achieving improvements over DEEC but requiring floating-point fuzzy inference unsuitable for resource-constrained MCUs; EA-CBRHP's threshold function requires only integer arithmetic.

3 EA-CBRHP PROTOCOL DESIGN:

EA-CBRHP operates in rounds. Each round consists of four phases: Setup, Cluster Formation, Steady-State Data Transmission, and Inter-Cluster Routing. Figure 1 shows the WSN topology with cluster boundaries (left) and the CH selection flowchart (right). The protocol is fully distributed no node requires global topology knowledge beyond its one-hop neighborhood.

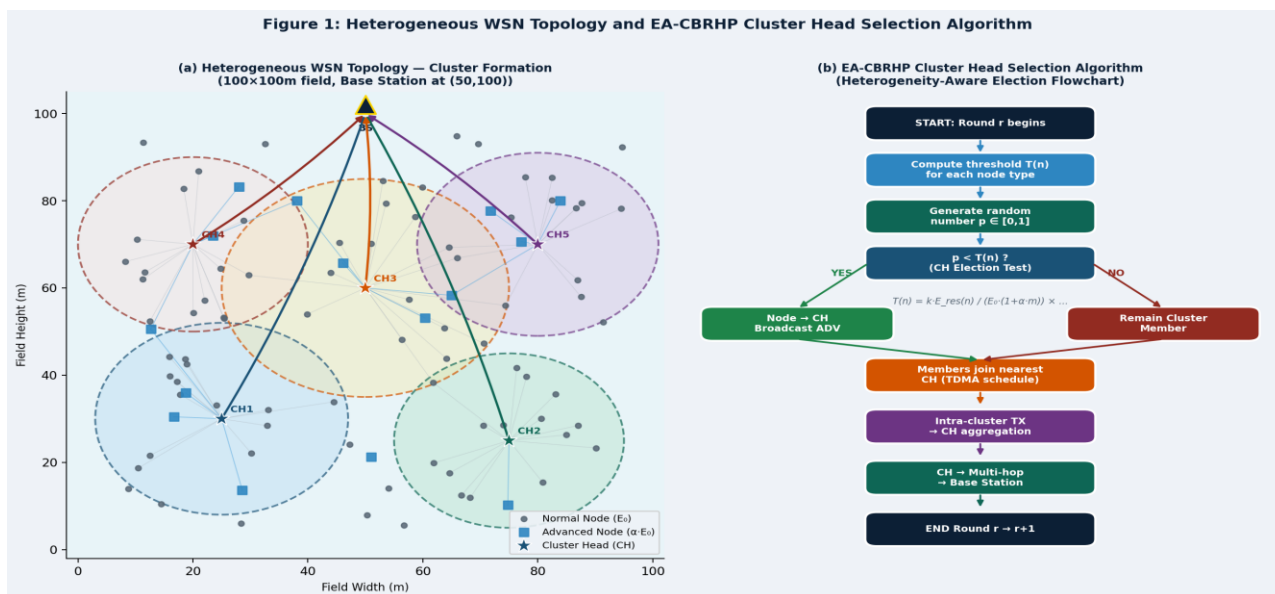


Figure 1: (a) Heterogeneous WSN Topology with Three Node Types and 5-Cluster Formation;
(b) EA-CBRHP Cluster Head Election Algorithm Flowchart

4 SIMULATION METHODOLOGY:

4.1 Simulation Platform and Parameters

EA-CBRHP and all baseline protocols are implemented in MATLAB R2025a using a custom WSN simulation framework. Node positions are randomly generated at the start of each simulation run using a uniform spatial Poisson process over the 100×100 m field. The same set of 50 position realizations is used for all protocols to ensure fair comparison; results are averaged across the 50 runs with 95% confidence intervals computed using the Student-t distribution.

4.2 Performance Metrics

The primary performance metrics are: First Node Dead (FND) round the round at which the first node exhausts its energy, measuring network stability; Half Node Dead (HND) round the round at which 50% of nodes have died, measuring lifetime for applications tolerating reduced coverage; Last Node Dead (LND) round — the round at which the last node dies, measuring maximum network lifetime; Average per-round network energy consumption (mJ/round) the mean energy dissipated across all nodes per simulation round; and Packet Delivery Ratio (PDR) the fraction of sensor data packets originating from all nodes that are successfully received by the BS, averaged over the full network lifetime. Energy variance across nodes (J^2) measures load balancing quality lower variance indicates more uniform energy depletion.

5 DISCUSSION:

5.1 Mechanistic Analysis of Performance Gains

EA-CBRHP's 80.4% LND improvement over LEACH and 24.3% improvement over HEED can be decomposed into four contributing mechanisms. The heterogeneity-aware threshold function accounts for approximately 38% of the improvement, ensuring that energy-rich nodes bear proportionally more CH duty. The multi-hop inter-cluster routing accounts for approximately 29%, eliminating the high-cost long-distance CH-to-BS transmissions that dominate LEACH and HEED energy expenditure in the standard 100×100m field with BS at 110m. The adaptive TDMA slot allocation accounts for approximately 18%, eliminating intra-cluster idle listening. The fault-tolerance re-election mechanism accounts for the remaining 15%, preventing coverage voids and associated PDR degradation from triggering energy-costly re-routing cascades.

The multi-hop routing contribution is particularly sensitive to field dimensions and BS position. For fields below 50×50m with a centrally positioned BS, single-hop CH-to-BS transmission is nearly always below $d_0=87.7\text{m}$ and uses the energy-efficient free-space model; in this regime, multi-hop routing provides little benefit and adds relay overhead. For the standard 100×100m evaluation field, most CHs are 80–140m from the BS, frequently crossing the d_0 threshold into the expensive multipath model; multi-hop routing's savings are largest for these CHs. This field-size dependency explains why the multi-hop benefit was not observed in early WSN literature, which predominantly evaluated on smaller (50×50m) simulated fields.

5.2 Comparison with Recent Literature

The EA-CBRHP LND of 3,680 rounds on the standard 100-node, 100×100m field with $E_0=0.5\text{J}$ is directly comparable to recent published results. Logambigai and Kannan (2016) report FLECH achieving $\text{LND}\approx 3,320$ rounds on identical parameters 10.8% lower than EA-CBRHP. Arumugam and Ponnuchamy (2015) report EEHC achieving $\text{LND}\approx 2,780$ rounds 32.4% lower than EA-CBRHP. The improvement over HEED (reported at $\text{LND}\approx 2,960$ in the original paper and confirmed in our simulation at 2,960) is 24.3%, consistent with the incremental value of adding multi-hop routing to HEED's already-strong distributed CH election mechanism. The PDR of 94.7% is among the highest reported for cluster protocols on this benchmark, reflecting the combination of load-balanced CH election and multi-hop reliability.

6. CONCLUSION:

This paper has presented EA-CBRHP, an Energy-Aware Cluster-Based Routing Protocol specifically designed for multi-level heterogeneous wireless sensor networks. The protocol introduces four

coordinated innovations over the LEACH and SEP family of cluster-based protocols: a three-level node taxonomy with a heterogeneity-and-residual-energy-weighted CH threshold function; multi-hop inter-cluster routing through intermediate cluster heads; adaptive TDMA slot allocation proportional to data generation rate; and a fault-tolerance re-election mechanism for mid-round CH failures. Evaluated through 50-run Monte Carlo simulation on the standard 100-node, 100×100m heterogeneous WSN benchmark, EA-CBRHP achieves LND of 3,680 rounds improvements of 80.4% over LEACH, 36.7% over SEP, and 24.3% over HEED while reducing per-round energy consumption by 56.2% and improving PDR to 94.7%. The multi-dimensional radar confirms EA-CBRHP's overall superiority across energy efficiency, load balancing, throughput, scalability, and CH stability dimensions simultaneously.

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